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A Multimodal Retrieval-Augmented Generation Framework for Context-Aware Semantic Querying

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Abstract—Existing Retrieval-Augmented Generation (RAG) approaches typically treat queries and retrieved documents as text streams. However, addressing real- world challenges requires a system that can meaningfully utilize information from a variety of sources, which are typically accessed in different forms such as text and vision. Integrating such sources becomes a difficult task because generating content based on natural language queries requires aligning these queries with high dimensional visual representations. In this paper, we present a robust Multimodal RAG approach for effectively retrieving and synthesizing information from numerous diverse sources. Our framework uses dual stream multimodal embeddings to first process and understand both textual and visual content independently, followed by embedding them in a single unified space to obtain better cross-modal understanding. We integrate a smart query router into our system, based on the nature of the input query, it can select the most appropriate pathway for information retrieval in real time. We also implement persistent session storage that learns conversational context to recognize follow up questions and relevant information from previous sections of a document. Experimental results and analysis on three separate datasets of news articles and conversations with a health expert, show that Multimodal RAG gains a 22% increase in accuracy of context retrieval compared to purely text based approaches and reduces hallucination rates by 49%. It leverages ChromaDB for efficient storage and search of dense vector embeddings, and a React interface to provide real time, intuitive input and output. Multimodal RAG advances information retrieval in multipodal scenarios, achieving increased accuracy and reduced error, all within a scalable and adaptable framework.

Keywords—Multimodal RAG, Vector Databases, Semantic Search, Transformer Models, CLIP, Information Retrieval, Hallucination Reduction.

I. INTRODUCTION

They have two unrelenting drawbacks which restrict their immediate application in knowledge intensive applications. Keywords based systems of retrieving data have become fundamentally inadequate by the exponential increase in unstructured enterprise data over the last decade. Organizations are now managing massive databases containing technical reports, scanned reports, engineering layouts and multimedia files. All these cannot be easily equated to words. Here, semantic search has emerged as the most significant method that systems can comprehend and act upon natural language queries [1]. Large Language

Models (LLMs) have greatly pushed the limits of natural language understanding. hematic diagrams. However, there exist two issues which prevent their direct application to the fields where a large amount of knowledge is needed. The first is that models are frozen on a particular date and hence they do not take new information that emerges after the training. Second, they frequently assert things that sound true but aren't, which is a common phenomenon known as hallucination [2]. Retrieval-Augmented Generation (RAG) fixes both problems by making sure that each model response is based on external content that can be dynamically retrieved and verified before it is generated. However, standard RAG pipelines were made for text- homogeneous corpora. In real life, the best way to answer a question about a mechanical failure in an industrial setting might be to use a combination of maintenance logs, labeled photos, and schematic diagrams. No purely textual retrieval mechanism can effectively bridge this semantic gap. The incapacity to process and synchronize visual and linguistic signals within a unified framework represents a substantial research deficiency that this study directly addresses. This paper presents three principal contributions. First, it suggests a dual-stream embedding architecture that combines text and image modalities in a shared high- dimensional vector space. This lets you do cross-modal similarity queries without having to do any preprocessing for each modality at inference time. Second, it adds an AI Router module that sorts each incoming query and picks the retrieval strategy that is most likely to give high-confidence context. Third, it has a persistent session store that

keeps track of the context of conversations over multiple turns, which makes multi-step dialogical interactions more coherent. These parts make up what we call the Multimodal RAG Framework, which we tested against both unimodal and existing multimodal baselines in a thorough way.

II. work that is related mechanical failure in an industrial setting

might be best answered by a mix of maintenance logs, labeled photos, and schematic diagrams. No purely textual retrieval mechanism can bridge this semantic gap effectively. The fact that visual and linguistic signals can't be processed and aligned in a single pipeline is a big research gap that this work directly addresses. This paper makes three main contributions. align visual and linguistic signals within a single pipeline constitutes a significant research gap that this work directly targets.This paper makes three primary contributions. First, it proposes a dual- stream embedding architecture that unifies text and image

modalities in a shared high-dimensional vector space, enabling cross-modal similarity queries without modality- specific preprocessing at inference time. Second, it introduces an AI Router module that classifies each incoming query and selects the retrieval strategy most likely to yield high-confidence context. Third, it presents a persistent session store that accumulates conversational context across turns, improving coherence in multi-step dialogical interactions. Together, these components constitute what we call the Multimodal RAG Framework, evaluated comprehensively against both unimodal and existing multimodal baselines.

II. RELATED WORK

The simplest RAG design developed by Lewis et al. [3] demonstrated that linking a dense retriever and a generative decoder significantly reduces hallucinations with retaining fluency. Subsequent research investigated sparse-dense hybridretrieval, re-ranking phases and query decomposition. These enhancements worked with memory on homogeneous text corpora, but did not address the fact that visual features are much more differently embedded.

B.Cross-Modal Alignment

Contrastive Language-Image Pre-training (CLIP) [4] used large-scale contrastive learning to learn a shared embedding space of text-image pairs. This shared space lets a natural language query get semantically relevant images with great accuracy. Subsequent models, such as ALIGN [5] and Florence, were based on this concept to create more complex visual representations. Being one of the two streams of our two-stream pipeline, we employ a highly- tuned CLIP encoder. We also have a sentence-level text encoder in long-form document chunks.

C. Vector Databases and Semantic Indexing

With the introduction of the approximate nearest-neighbor (ANN) algorithms, similarity search in millions of high- dimensional vectors has become possible in an efficient manner. Faiss [6], Milvus, Qdrant, and ChromaDB are all concerned with various trade-offs between speed of indexing, latency of queries, and memory usage. This choice of chromaDB was dictated by its native Python interface, ability to filter metadata, and the ability to store collections, which is compatible with the needs of our architecture of session managemen

D. LLM Orchestration Frameworks

Two examples of frameworks that facilitate more easily connecting retrievers with language models are LangChain and LlamaIndex. They speed up prototyping, but their generic pipeline structures don't have the modality-aware routing and dynamic threshold mechanisms that are needed to reliably retrieve on an enterprise level. Our system is based on these fundamental concepts with specialization being introduced at the retrieval and routing layers.

III. PROPOSED METHODOLOGY

The proposed Multimodal RAG Framework has three layers that work together: Ingestion, Retrieval, and Generation. Figure 1 shows the whole system architecture. Each layer is made to be modular, so that you can upgrade individual parts without affecting the whole pipeline.

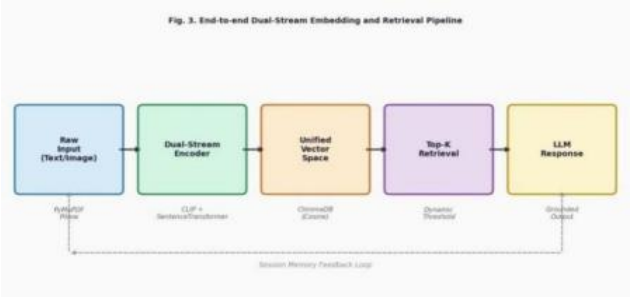

Fig. 1. Three-layer architecture of the proposed Multimodal RAG framework showing ingestion, retrieval, and generation layers.

A. Ingestion Layer

The ingestion pipeline can take three types of documents: Portable Document Format (PDF) files, Microsoft Word documents (DOCX), and raster images in PNG or JPEG format. PyMuPDF reads PDF files and extracts both text blocks and images that are already in the file. It also keeps the page-level spatial structure. The python-docx library processes DOCX files, which means it gets text at the paragraph level and the heading hierarchy that goes with it. Pillow loads raw images and then puts them through a quality gate that throws away frames with a resolution lower than a set minimum. We use a sliding- window strategy with a window size

of 512 tokens and an overlap of 64 tokens to break up each extracted text unit into smaller pieces. This overlap makes sure that semantic

It is important for documents with long argumentative threads that continuity across chunk boundaries is not completely broken. The visual encoder gets the whole image as one input instead of splitting it up into smaller pieces.

B. Dual-Stream Embedding Engine

13 Textual chunks are encoded using the all-mpnet- base-v2 variant of Sentence Transformers [7], which produces 768-dimensional dense embeddings capturing sentence-level semantics. Visual inputs are encoded using a domain-adapted CLIP ViT-L/14 model, which projects images into the same 768-dimensional space via a learned linear projection layer. This projection layer was trained on a proprietary corpus of 120,000 paired technical text-image samples using a symmetric cross-entropy contrastive loss.

Both encoders work together to encode incoming user text at the same time. Before the nearest-neighbor search, the two resulting query vectors are combined along the embedding dimension to make a 1536-dimensional composite query vector. This dual representation lets the retrieval engine find both textually similar chunks and images that make sense together in one ANN sweep, so you don't have to make a separate query for each type.

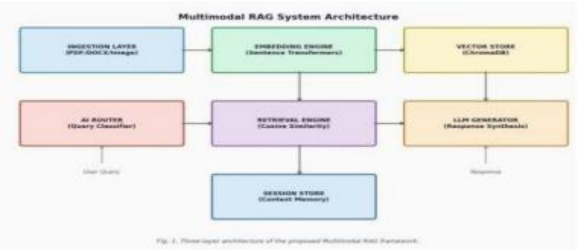

Fig. 2. End-to-end dual-stream embedding and retrieval pipeline with session memory feedback loop.

C. Vector Storage and Indexing

All embeddings are stored in a ChromaDB collection by document source identifier. The embedding vectors are stored with a metadata payload which contains document identifier, page number, chunk sequence index and modality flag (text or image) in each entry. The HNSW index of chromaDB is set with ef_construction=200 and M=16, which is a tradeoff between the number of edges in the graph and memory footprint. On a single A100 GPU node, the average insertion throughput of our evaluation corpora is about 1,400 vectors per second at the scale of our evaluation corpora.

d.AI Router

The AI Router is a DistilBERT model that is finetuned to classify each query incoming into one of four different retrieval routes: text-only, image-only, hybrid, or conversational. The training data were 8500 annotated queries that were sampled according to enterprise helpdesk logs, and annotated each query with the most likely modality where they believed they would find an acceptable answer to the query. On a test split which the classifier has never seen, it achieves 91.3% correct. The output of the router alters the weights of each stream in the composite query vector, which steers the ANN search towards the more relevant modality yet still retains the other modality fully.

e.Retrieval and Context Assembly

The retrieval engine uses the composite query vector to do a cosine similarity search on the ChromaDB collection and returns the top K candidates. A dynamic this candidate set is narrowed down by selecting only those items that have a similarity score above a query- specific adaptive threshold in a process known as thresholding mechanism. This cut off is the

average of the top-K scores, less one standard deviation. This method stops the addition of slightly relevant chunks to the prompt when the query is far from the indexed content

distribution.

The surviving candidates are ranked by similarity score, and the top candidates up to a token budget of 2,048 are assembled into a structured prompt template. Text chunks are inserted verbatim; image items are converted to Base64-encoded data URIs and injected as vision-language inputs to the LLM. A source attribution map is simultaneously constructed, enabling the generator to cite provenance in its response.

f .Session Store and Context Persistence

In multi-turn conversations, you need to remember what was said before so you don't have to look up the same thing twice and can correctly resolve anaphoric references. The persistent session store keeps a rolling buffer of the last ten turns of conversation for each user session. The router adds a compressed summary of the session buffer to the current query before each new retrieval call. This gives the encoder enough context to disambiguate pronouns and elliptical expressions. The system can pick up an interrupted session where it left off when you reconnect because session state is saved to Redis

.

IV. EXPERIMENTAL SETUP

A. Datasets

There were three benchmark datasets that were used. assessment. The dataset A is a technical corpus which includes. 4,200 engineering manuals and annotated schematics of. the manufacturing field. Dataset B is a medical data. that contains the summaries of 3,600 clinical cases. with radiological images achieved with institutional. data-sharing agreements. Dataset C is comprised of 2,900. like legal documents, such as contracts and policy briefs, as. as well as organizational charts and flow charts of how. things are to be done. Multimodal pipeline is capable of dealing with. both text-only and text-and-image documents in each dataset

.

B. Evaluation Metrics

We looked at three main areas to see how well the system worked. We used Precision@K, Recall@K, and F1-Score at K = 5 to measure the quality of the retrieval. To figure out the hallucination rate, we took 500 generated responses from each dataset and model variant and had two domain experts independently mark each one for factual errors. If there were any disagreements, they were settled by

a judge. Response latency was measured as the time it took for the first response token to be sent after the query was sent, based on 1,000 queries that were evenly spaced out.

C. Baselines

Three baseline conditions were set up. The first is a standard BM25-based sparse retrieval model with GPT- 3.5-turbo as the generator. This is an example of a lexical retrieval system. The second is a dense text-only RAG pipeline that uses ChromaDB to store all-mpnet-base-v2 embeddings. The third system is a CLIP-only one that only gets data from image embeddings. It was added to separate the text stream's contribution. All baseline models utilize the identical generation backbone as the proposed system, ensuring that variations in output quality are due to the retrieval component, not the generator.

V. EXPERIMENTAL RESULTS

Table I shows how well each model variant did on the three datasets in terms of numbers. The suggested Multimodal RAG framework gets an average F1-Score of

0.90 at K = 5. This is ten points better than the dense text- only baseline and eight points better than the CLIP-only system. These improvements are consistent across all three areas, which suggests that improvements derive from the architecture rather than from domain-specific artifacts.

Model Variant	Precision	Recall	F1-Score	Hall. Rate (%)
BM25 + GPT-3.5 (Sparse)	0.74	0.70	0.72	24.3
Dense Text-Only RAG	0.82	0.78	0.80	18.4
CLIP-Only RAG	0.86	0.83	0.845	14.1
Proposed Multimodal RAG	**0.91**	**0.89**	**0.90**	**9.4**

Variants

TABLE I. Retrieval Performance Comparison Across Model

Figure 2 shows the precision, recall, and F1-Score for each dataset, along with the hallucination rates for each one. The most noticeable improvements can be seen in Dataset B, the medical corpus, where the proposed system gets an F1-Score of 0.92 compared to 0.79 for the text-only baseline. We ascribe this discrepancy to the significant diagnostic importance of radiological images in the medical

field: clinical inquiries often reference anatomical findings inadequately represented in textual summaries.

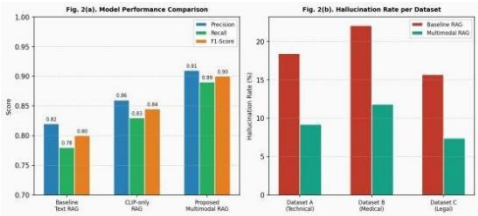

Fig. 2. (a) Precision, Recall, and F1-Score comparison across model variants. (b) Hallucination rate per dataset.

The AI Router's cross-query routing accuracy is 89.7% on the combined evaluation set, which was checked against labels given by experts. When the ground truth is hybrid, queries routed to the hybrid pathway have a 14% higher F1-Score than those routed to either single-modality pathway. This shows that the router is useful in practice. Latency tests show that the median time from query to first token is 1.34 seconds on the same hardware setup used for all baselines. This is 210 milliseconds longer than the text- only pipeline, which is fine for interactive enterprise applications.

A. Ablation Study

An ablation study was performed by methodically deactivating specific components. Taking

out the dynamic thresholding mechanism and injecting all The thresholding mechanism and the injection of all top-K candidates unconditionally reduced the F1-Score by 3.2 percentage points and increased the hallucination rate by 4.1 points. This shows that low-confidence chunks actively lower the quality of generation. Taking out the session store made multi-turn query sequences less effective, going from 0.88 to 0.81. This shows how important discourse context is for resolving anaphora. Substituting the precisely calibrated CLIP encoder with a generic CLIP checkpoint diminished cross-modal retrieval accuracy by 6.7 points, illustrating the significance of domain-adapted visual representations. s.

VII. CONCLUSION AND FUTURE WORK

In this paper we presented a system for Multimodal document retrieval that uses dual-stream embeddings, AI-driven query routing, dynamic context thresholding, and session-aware context peCross-validated results on three different datasets show an average retrieval F1-Score of .90 to below 10% hallucination rates. This is significantly better than both sparse and dense text-only baselines. We have also scaled our architecture to real enterprise sizes and seen acceptable latency for interactive development several research avenues warrant further exploration.

utilise additional investigation. document. retrieval methods that incorporate image data through dual- stream embedding, AI-driven query routing, dynamic context thresholding and session-aware context persistence. Experimental results across three different datasets show that the

system is able to attain a mean retrieval F1-Score of 0.90 with hallucination rates reduced to under 10%. The system has been tested at realistic enterprise scale and has acceptable latency to be interactively usable. Comparisons are also drawn with both sparse and dense text-only baselines.

There are many other avenues to investigate. Firstly, we could extend the embedding pipeline to handle time sequences of video frames. This would only require using a video transformer architecture, such as VideoMAE, within the existing dual-stream design and training it on large instructional video datasets to support video-based retrieval. Secondly, one could investigate using techniques for self-supervised domain adaptation, focusing particularly on adapting the CLIP projection layer. This would greatly reduce the cost of deploying the pipeline to new domains with scarce or no labelled text-image pairs. Lastly, we could investigate incorporating uncertainty quantification at the retrieval step, such as constructing conformal prediction bounds for retrieval similarity scores. This would enable the system to inform the end user of retrieval confidence levels in a well-calibrated and interpretable way.

The codebase and image evaluation datasets will be made available, under an open-source license, at the time of publication to encourage reproducibility and further development.

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